

Integration, implementation and reporting outlooks of sustainability in higher education institutions (HEIs): index and case base validation

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Abstract

Purpose – The integration of sustainability is more matured in the corporate sector. However, the topic received less attention in higher education institutions (HEIs). Therefore, this study aims to conceptualize an integration framework and proposes sustainability integration and implementation index in the HEIs.

Design/methodology/approach – The index was developed from previous literature. For the implementation of the index, three universities were selected. The required data collected from websites, internal documents and interviews with the high officials of these universities.

Findings – Sample universities have somehow inclination toward the broad agenda of sustainable development, but still, they lack integration, implementation and reporting of their sustainable practices. Moreover, the departments have positive and practical intent toward sustainability especially regarding the environment, designing curriculum, teaching, research activities and volunteerism. However, in assessment, the social and economic contributions of these universities revealed to be somewhat mild. The findings also evidenced heterogeneity among the sample institutions to follow the said agenda which reflects a lack of awareness and policy or a centralized or regulatory push from the top.



Research limitations/implications – The study is limited to three universities in Khyber Pakhtunkhwa, Pakistan. Hence, in the future, the framework and index may be tested in other universities of the country or region. Nevertheless, the study carries theoretical significance in the literature of sustainability.

Practical implications – The study has practical implications for academia, regulatory bodies and universities to integrate, implement and report sustainability in HEIs.

Originality/value – The study proposes and validates the sustainability integration and implementation index in HEIs. The index is unique and has originality or value particularly for HEIs of the developing countries.

Keywords Education for sustainable development, Integration and implementation, Higher education institutions, Index, Pakistan

Paper type Research paper

Introduction

In the past few decades, sustainability has become a buzzword or concept that grabbed much of the attention in almost every field of life. Among others, this has also changed the landscape of higher education institutions (HEIs) to extend support and address the emerging challenges of sustainability (Findler *et al.*, 2019). Besides imparting the education and carrying out research in the field of sustainability, HEIs are also supposed to set an example to integrate sustainability in their practices and operations (on and off-campus). To do so, many HEIs around the globe have formulated and implemented several sustainability-related initiatives on their own and voluntary basis. The HEIs have somehow integrated sustainability by initiating and developing their teaching, curriculum, research and scholarship, campus operations, community outreach, financial management and everyday activities (Alshuwaikhat *et al.*, 2016; Sonetti *et al.*, 2016). Simply, these HEIs are adhering to the broad agenda of sustainable development, as defined by the Brundtland Commission Report 1987 that the development that “meets the need of the present generation without compromising the ability of future generations to meet their needs” (WCED, 1987).

In doing so, HEIs have developed and used numerous tools to integrate, implement and report their sustainability-related practices and activities (Alghamdi *et al.*, 2017; Schoormann *et al.*, 2017). Some of these tools include the Sustainability Assessment Questionnaire (SAQ) which addresses curriculum, research, operations, faculty development, public and student engagement, and administration dimensions of HEIs for evaluating their sustainability. Likewise, the Sustainability Tracking Assessment and Rating System (STARS) adds innovation dimension to the SAQ. Association for Advancement of Sustainability in Higher Education (AASHE) has, however, focused on limited dimensions as compared to SAQ and STARS as it includes all dimensions except for faculty development, students’ engagement, administration and innovation). Benchmark Indicator Questions alternative university appraisal (BIQ-AUA) is another tool that focuses on four dimensions, i.e. Red de Ciencia, Tecnologia, Innovacion, Educacion. Besides, Ambienalem Iberoamerica (CITE-AMB) accounts for four, whereas both the Adaptable Model for Assessing Sustainability (AMAS) and Graz Model for Integrative Development (GMID) address five dimensions. Graphical Assessment of Sustainability in Universities (GASU) and Sustainable Campus Assessment System (SCAS) covers six dimensions (introducing reporting as a dimension). Besides, the Sustainable University Model (SUM) and Unit-based Sustainable Assessment Tool (USAT) also focuses on four basic dimensions.

Despite these numerous available tools for the HEIs, many universities in the developing world, especially in South East Asia are still not following and integrating sustainability practices. In some cases, these universities follow their measures to integrate sustainability into routine campus operations. Because of this, it could be inferred that sustainability in HEIs is encountered with numerous issues in developing countries which include lack of institutional support and governance, among others (Filho *et al.*, 2015). Policy failure and lack of commitment,

readiness, consensus, guidelines, planning, leadership, infrastructure, resources, innovation, technology and training and development are those other factors responsible for the failure of sustainability in the HEIs of developing countries. Quality education being a crucial part of 2030 sustainable development agenda of World Bank and Sustainable Development Goal 4 (SDG4) (The World Bank Group, 2019), the HEIs in developing countries are yet to incorporate sustainability in their culture, curriculum, operations and planning (Adams *et al.*, 2018).

Likewise, very limited research work carried out to explore and assess the process of sustainable development in HEIs of the developing countries. More specifically, there is a tiny amount of research conducted in HEIs of the developing countries (Alghamdi *et al.*, 2017) as only a few studies conducted and published like Naeem and Neal (2012) for Asia-Pacific, Alghamdi *et al.* (2017) for Saudi Arabia, Cronemberger de Araújo Góes and Magrini (2016) in Brazil and Zahid *et al.* (2016) in Malaysia, among others. Given this, there is no comprehensive study on specific aspects of sustainability in HEIs of developing countries. Thus, there is a genuine need and space to investigate the sustainability practices in HEIs and that too in developing countries where, as compared to developed countries, the concept is yet emerging and in embryonic stages even at the corporate level. Subsequently, this study mainly aims to answer how the universities of a developing country like Pakistan integrate sustainability into their routine campus operations, curriculum development and community outreach to satisfy the demand of multiple stakeholders, and what would be the way forward for these universities in Pakistan. Pakistan being a developing country has been struggling with several issues since its independence in 1947. Political and economic instability, security issues, lack of leadership and many other challenges like the war against terrorism and post 9/11 scenario has kept the country especially its North-west part – the province of Khyber Pakhtunkhwa (KP) – away from the desired development. The education and particularly higher education in Pakistan, despite the increase in enrollment that reached to the historical high level of over 1.6 million with the teaching faculty exceeding 56,000, is still struggling in regard with quality that makes the case worthy, appealing and significant for investigation.

To answer the research question, this study investigates sustainability in three HEIs of Pakistan, i.e. two private universities – City University of Science and Information Technology (CUSIT) and Sarhad University of Science and Information Technology (SUIT) and one public sector universities – University of Science and Technology Bannu (USTB) located in the KP province. The study would contribute many implications for the multiple stakeholders in Pakistan. The study identifies the most followed sustainability indicators in the HEIs. Also, the study updates the top management of HEIs to implement and integrate sustainability for the satisfaction of multiple stakeholders. Besides, the study also combines these indicators in the sustainability index followed by the HEIs which will assist them in integrating sustainability into their activities and operations. The study has significant implications for the policymakers and regulatory bodies such as the Higher Education Commission (HEC) of Pakistan, especially regarding the green operations and campuses of the HEIs. In connection with the theoretical contribution, the study adds new insights to enrich the extant prior literature of education for sustainable development (ESD) particularly in the context of developing countries. Last but not least, the study applies the crux of the stakeholder theory (Donaldson and Preston, 1995) in that HEIs are responsible to manage the demands of multiple stakeholders in different dimensions of ESD.

The remaining of this paper is structured as a literature review, methods, findings, conclusion, recommendations and way forward.

Literature review

Sustainability in higher education institutions

As the concept of sustainability has been adopted by corporations, they have started sustainability reporting (SR). However, still, the concept is quite naive and even has no

operational definition in terms of reviewing the operations and strategies of HEIs ([del Mar Alonso-Almeida et al., 2015](#)). It has been noted that after the Stockholm Conference in 1972, education was formally recognized on an international level to play an important role in promoting environmental protection and conservation. Since then, more HEIs are engaged in embedding environmental education and ESD into their system core activities like education, research, campus operations, community outreach, assessment, and reporting. [Table 1](#) shows some widely recognized higher ESD (HESD). It is argued that sustainability practices of the HEIs may not involve SR, but there may be numerous other tools and techniques developed and used around the globe to monitor practices related to sustainability in HEIs.

Sustainability assessment tools

To assess and report the sustainable practices in HIEs, many sustainability tools are developed, implemented/practiced, revised and modified. To evaluate and assess the sustainable development related initiatives and activities of HEIs, it is important to know about Sustainability Assessment Tools (SATs) and their role which has gained much attention these days ([Lambrechts, 2015](#)). These tools include the SAQAASHE, Campus Sustainability Assessment Framework (SCAF) and STARS, among others, ([Berzosa et al., 2017](#)) as reported in [Table 2](#).

The SAQ covers the dimensions of research, operations, curriculum, public management, faculty development, administration and student engagements. The role of AASHE is limited as compared to the dimensions of STARS and SAQ as STARS, in addition to SAQ, also accounts for innovations. Likewise, SUM considers four dimensions (education, research, outreach and SR), and USAT deals with four dimensions (education, research, operations, and outreach). The BIQ-AUA also accounts for four main dimensions. Another model to assess sustainability in HIEs is the AMAS that emphasizes five dimensions. Also, the focus of the GMID is on five dimensions and the GASU model focuses on six dimensions. Similarly, the SCAS considers six dimensions that include reporting as a new dimension.

The tools, as discussed, are different from each other, especially concerning their applicability, as these are developed and designed for various areas. Thus, based on the differences in applicability and assessment areas of SATs, no universal or standard tool could help HEIs to set directions for sustainable development in the future ([Lozano et al., 2015](#)). Among others, this could be the one plausible rationale for no or low sustainability in HIEs and also the reason for not following the path of sustainable development.

Researchers have developed and analyzed various SATs for HEIs, but none of the recent studies have emphasized a single tool. Over the years, an increase in the integration of SATs has been noted and researchers are focusing on various combinations of different SATs. In this strand, [Shriberg \(2002\)](#) used 11 tools, [Yarime and Tanaka \(2012\)](#) used 16, [Alghamdi et al. \(2017\)](#) emphasized on 12 and [Findler et al. \(2019\)](#) used 19 tools. These SATs have never been free of limitations; thus, researchers have integrated and adapted these assessment tools as per requirement. For example, [Cronemberger de Araújo Góes and Magrini \(2016\)](#) adapted SAT in the HEIs of Brazil, [Alshuwaikhat et al. \(2016\)](#) used the same approach in the case of Saudi Arabia, [Aleixo et al. \(2016\)](#) in Portugal, and [Zahid et al. \(2016\)](#) in Malaysia. The most commonly adapted SATs in integrated studies with a focus to assess the HEI's education, operations, research, outreach and administration internally are SAQ, AASHE, USAT and STARS.

The traits of education, economic, social and environmental outcomes are the main concerns of all the major SATs ([Findler et al., 2019](#)). The objective of the most common SATs is to focus on the ranking and self-assessment but their methods, scope and the

Table 1.
HIEs developed and
signed a collection of
declarations, charters

Year	Event/declaration	Level or focus
1972	Stockholm Declaration on the Human Environment, United Nations Conference on the Human Environment, Sweden	Society
1975	The Belgrade Charter, Belgrade Conference on Environmental Education, Yugoslavia Tbilisi	Education
1977	Tbilisi Declaration, Intergovernmental Conference on Environmental Education, Georgia	Education
1987	"Our Common Future", The Brundtland Report	Society
1990	Talloires Declaration, Presidents Conference, France	Higher Education
1991	Halifax Declaration, Conference on University Action for Sustainable Development, Canada	Higher Education
1992	Report of the United Nations Conference on Environment and Development (Rio Conference), Agenda 21, Chapter 36: Promoting Education, Public Awareness and Training and Chapter 35: Science for Sustainable Development	Education
1992	Association of University Leaders for a Sustainable Future, founded, USA Kyoto	Society
1993	Kyoto Declaration, International Association of Universities Ninth Round Table, Japan	Higher Education
1993	Swansea Declaration, Association of Commonwealth Universities' Fifteenth Quinquennial Conference, Wales	Higher Education
1993	COPERNICUS University Charter, Conference of European Rectors (CRE)	Higher Education
1996	Ball State University Greening of the Campus conferences were in 1997, 1999, 2001, 2003, 2005, 2007, and 2009	Higher Education
1997	Thessaloniki Declaration, International Conference on Environment and Society: Education and Public Awareness for Sustainability, Greece	Higher Education
1999	Environmental Management for Sustainable Universities (EMSU) conference, first held in Sweden. Following conferences in 2002 (South Africa), 2004 (Mexico), 2006 (U.S.A.), 2008 (Spain), and in 2010 in The Netherlands	Higher Education
2000	Millennium Development Goals The Earth Charter Global Higher Education for Sustainability Partnership (GHESP)	Society Society Higher Education
2001	Lüneburg Declaration on Higher Education for Sustainable Development, Germany	Higher Education

(continued)

Year	Event/declaration	Level or focus
2002	World Summit on Sustainable Development in Johannesburg, South Africa (Type 1 outcome: Decade of Education for Sustainable Development; Civil Society outcome: the Ubuntu Declaration)	Society
2004	Declaration of Barcelona	Higher Education
2005	Start of the UN Decade of Education for Sustainable Development (DESD)	Higher Education
	Graz Declaration on Committing Universities to Sustainable Development, Austria	Higher Education
2009	Abuja Declaration on Sustainable Development in Africa: The role of higher education in SD, Nigeria	Higher Education
	Torino (Turin) Declaration on Education and Research for Sustainable and Responsible Development, Italy	Higher Education
2012	Rio 20 Higher Education Sustainability Initiative, Brazil	Higher Education

Table 1.

Table 2.
Sustainability
assessment tools and
dimensions

Dimensions	SAQ	STARS	AASHE	BIQ-AUA	CTTE-AMB	AMAS	GASU	GMID	GP	SCAS	GMID	SUM	USAT
Education/Curriculum	X	X	X	X	X	X	X	X	X	X	X	X	X
Research	X	X	X	X	X	X	X	X	X	X	X	X	X
Operations	X	X	X	-	X	-	X	-	X	-	-	-	-
Faculty development	X	X	-	-	-	-	X	-	X	-	-	-	-
Outreach/Public engagement	X	X	X	X	X	X	X	X	X	X	X	X	X
Student engagement	X	X	-	-	-	-	X	X	X	-	X	-	-
Planning and administration	X	X	-	X	-	X	X	X	X	X	X	-	-
Innovation	-	X	-	-	-	-	-	-	-	-	-	-	-
Identity	-	-	X	-	-	-	-	-	-	-	-	-	-
Sustainability in campus/SR	-	-	-	-	-	-	-	-	-	X	-	X	-

Notes: SAQ = Sustainability Assessment Questionnaire; STARS = Sustainability Tracking Assessment and Rating System; AASHE = Association for Advancement of Sustainability in Higher Education; BIQ-AUA = Benchmark Indicator Questions alternative university appraisal; and CTTE-AMB = Red de Ciencia, Tecnologia, Innovacion, Educacion. Besides, Ambienalem Iberoamerica; AMAS = Adaptable Model for Assessing Sustainability; GASU = Graphical Assessment of Sustainability in Universities; GMID = Graz Model for Integrative Development; SCAS = Sustainable Campus Assessment System; GMID = Graz Model for Integrative Development; SUM = Sustainable University Model; and USAT = Unit-based Sustainable Assessment Tool; X = Dimension covers; - = Dimension does not cover

Source: Authors compilation

questions they carry are different. For example, SAQ aims to assess HEIs, AASHE's role is focused on self-assessment, USAT is concerned with self-assessment and STARS is related to self-assessment based on developed criteria for HIEs' sustainability. Further details of these SATs and dimensions have been reported in different previous studies (Alghamdi *et al.*, 2017; Berzosa *et al.*, 2017; Findler *et al.*, 2019; Lozano *et al.*, 2015; Yarime and Tanaka, 2012).

Besides, prior studies have focused on quantitative, qualitative or mixed methods of research in their endeavors. Most of them preferred mixed-method and both the open- and close-ended questions for their inquiry (Alghamdi *et al.*, 2017; Yarime and Tanaka, 2012). It is also evident that the main focus of these studies was to compare different SATs and indicators, single case assessment, development of the framework, content analysis or SR, among others. Though substantial research work has been carried out regarding the development, reporting and ranking mechanism, examination of assessment tools but still it noted that there is a space for further exploration under different SATs. Hence, there is no consensus about any single tool to be used for evaluating the sustainability of HEIs.

An overview of the higher education commission and higher education institutions in Pakistan

Like other countries around the world, a tremendous increase has been noted in the number of HEIs or degree awarding institutes in the recent past in Pakistan. The number of recognized HEIs in Pakistan has reached to 195 in 2019. It is also observed that the number of private institutions has increased over time and at present, there are 79 private HEIs in Pakistan (Rizwan *et al.*, 2016). Besides, the enrollment of students in these universities has also been increased which is a sign of trust over the private universities for imparting quality education an increase has been (Halai, 2013). To cope with the increased enrolment and other emerging challenges in today's academia, there is a dire need to attract, engage and retain competent faculty and provide all the required facilities to spur knowledge and research.

In the case of Pakistan, it is observed that HEIs are lagging especially concerning a standard curriculum, student's motivation and high standards to ensure good quality of education and research. Moreover, the deficiency of highly qualified and competent faculty in some cases and a weak intention or attention toward knowledge economy along with a low budget for HEIs are also the problems in academia at a higher level. Connected to this a large pool of talent is affected or even deprived sometimes because of higher fee structure and low financial support for needy students in the HEIs (Farid *et al.*, 2015; Qureshi *et al.*, 2012). In this regard, the role and efforts of HEC Pakistan cannot be overlooked in that to enhance the quality of higher education, curriculum and research as the commission have established numerous accreditation councils to work closely with specific professional bodies. The establishment of Quality Enhancement Cells (QEC) and Quality Assurance Agency (QAA) are others endeavor of the commission which reflects its commitment toward consistent monitoring and uniform policies for uplifting the quality of higher education in the country.

To make sure the compliance of HEIs, HEC has also established independent QEC under QAA in each HEI of the country. Based on the institutional performance, HEC ranks all universities (both the public and private sector) in four levels, i.e. W, X, Y and Z (where W is the top and Z is the worst level or category) in Pakistan. HEC has also designed the process of Institutional Performance Evaluation (IPE) to evaluate the performance of HEIs under the Medium-Term Development Goals framework 2002-2007. Furthermore, an established criterion composed of 11 standards (Table 3) is also used to evaluate the performance of

universities. However, the 11 standards complimented by 144 evidenced-based questions do not fully qualify or address the true essence of sustainable development. IPE is an overall assessment and ranking system of HEIs; the self-assessment process under the supervision of the QEC department for each program is separately conducted and reported to QAA. The self-assessment report (SAR) is composed of eight standards as given in Table 4.

To maintain the level of student learning, academic standards, to provide linkage to the goals and mission of HEIs, the process of self-assessment is carried out in each program at university. The self-assessment process is composed of eight criteria achieved through 31 questions or standards having nothing with sustainable practices or sustainability which raises a question over the role and seriousness of the concerned authorities in the ranking or assessment criteria. However, a glimpse of sustainable practices can be seen in the standards of accreditation councils. For example, the standards of the Pakistan Engineering Council are according to the Washington Accord www.ieagreements.org/accords/washington/signatories/.

The SAR and IPE programs of QAA under the supervision of HEC can be used to achieve the fundamentals of SATs (education, operations, education, administration and outreach). The IPE program has the potential to cover all the internal factors. The buzzword of sustainability has entered the corridors of HEC and following the sustainable development goals (SDGs), governing boards and chairman (HEC) have now focused on sustainability. Representatives of 15 HEIs and HEC have shown their

Table 3.
Institutional
performance
evaluation (IPE)
standards and major
areas

Standards	Major area of evaluation
Standard 1	Mission statement and goals
Standard 2	Planning and evaluation
Standard 3	Organization and evaluation
Standard 4	Integrity
Standard 5	Faculty
Standard 6	Students
Standard 7	Institutional resources
Standard 8	Academic programs and curricula
Standard 9	Public disclosure and transparency
Standard 10	Assessment and quality assurance
Standard 11	Student support service
Source: Authors compilation	

Table 4.
Self-assessment
report (SAR) criterion
for programs offered

Criteria	Major area of evaluation
Criterion 1	Program mission objectives and outcomes
Criterion 2	Curriculum design and organization
Criterion 3	Laboratories and computing facilities
Criterion 4	Student support and guidance
Criterion 5	Faculty
Criterion 6	Process control
Criterion 7	Institutional facilities
Criterion 8	Institutional support
Source: Authors compilation	

commitments to consider and include Green Metric in the assessment and ranking process of HEIs in 2018. The adoption and implementation of the green framework of the Universitas Indonesia model by the commission is an initiative that reflects its positive intent toward the subject. The intention of HEC toward sustainability assessment in HEIs is an encouraging sign, although it is yet in an infancy stage. In this regard, the use of the Green Metric to encourage students to opt-in the HEI's initiatives and to rank and rate universities is on the agenda of HEC. The focus of the Green Metric is to observe the indicators of environmental sustainability (energy use, transport, infrastructure, education and water consumption). However, some of the vital dynamics including economic and social indicators could not be covered under this ranking tool. Given this, an integrated and aggregate sustainability assessment approach is the call of the day that can cover all aspects of sustainability.

Research design

The case in point, that is selected for the present study is universities in Pakistan to assess the integration of sustainability into their routine operations, curriculum and outreach. The study purposely selected three universities in Khyber Pakhtunkhwa (KP), i.e. CUSIT, SUIT and USTB. The former two are private universities located in the provincial capital of KP, i.e. Peshawar, whereas the latter is a public sector university located in the remote southern district of KP - Bannu. The selection of universities has a rationale that the corresponding author of the study who initiated the idea invited many faculty members for research collaboration from different universities. However, only a few responded and expressed their willingness to peruse the task. Among the respondents, the study selected the two authors from other universities as there were no legal or organizational issues and both of them held senior positions in academics and administration to have or have easy access to the required information and say or value in the decision making. As discussed earlier, there is no regulatory pressure and consolidated efforts from the top management of the universities and hence, the sustainable development initiatives in both the private and public sector HEIs are not uniform in the country. One of the key limitations is that the study could include only three universities.

The data is collected from the websites and internal documents of these universities. For data collection, the study adapts universities' sustainable development initiatives checklist/index from the previous literature (Lozano *et al.*, 2014; Zahid *et al.*, 2016) exhibited in Table 5. This index covers integration, environment, curricula, research, social and economic development, awareness and volunteerism, assessment and reporting and infrastructure. To ground sustainable development initiatives, the study uses a qualitative content analysis procedure that is appropriate for both the inductive and deductive nature of research. In both cases, the qualitative content analysis processes involve three main phases such as preparation, organization and reporting of the results. The preparation stage consists of collecting the relevant data for the content analysis, making sense of the data and selecting the unit of analysis. The organization of the data involves categorization matrix development and record in the relevant section. For this purpose, "Yes" reported if the particular content existed and put "No" otherwise like many previous studies such as Findler *et al.* (2013) and Zahid *et al.* (2016). In the final phase, the results recorded and discussed by the content of the categories (Elo *et al.*, 2014). Many previous studies similar in nature applied the same methodology (Ceulemans *et al.*, 2015; Schoormann *et al.*, 2017; Zahid *et al.*, 2016). Applying the qualitative content analysis procedure, the data was collected from the concerned university's main websites, embedded website links, university internal

Table 5.
Education for
sustainable
development priority
areas

Initiative for sustainable development	USTB	CUSIT	SUIT
<i>Integrations</i>			
Top management commitment	Yes	Yes	Yes
SD included in the vision and mission, goals and objectives	Yes	Yes	Yes
Policy for implementing SD	Yes	Yes	Yes
Strategic plan for implementing sustainability	Yes	Yes	Yes
Targets and benchmarking	Yes	Yes	Yes
Stakeholders identification	Yes	Yes	Yes
Staff dedicated to SD	Yes	Yes	No
SD working group	No	Yes	No
SD office	No	Yes	No
University leaders provide an SD budget or in-kind support	No	Yes	Yes
<i>Environmental sustainability</i>			
Plans to improve energy efficiency	Yes	Yes	Yes
Energy-efficient equipment	Yes	Yes	Yes
Renewable energy usage	No	No	No
Plans to reduce GHG emission	No	Yes	Yes
Efforts to reduce GHG	Yes	Yes	No
Plans to improve the prevention and management of waste	No	No	No
Waste bins to separate and recycle waste	No	Yes	Yes
Policies and activities to reduce paper consumption	No	Yes	No
Plans for water and wastewater management	Yes	Yes	Yes
<i>Curricula</i>			
Course on SD for all students	Yes	Yes	Yes
Optional courses on SD in the curriculum	Yes	Yes	Yes
Integrated sustainability courses throughout all curricula	Yes	Yes	Yes
Invited lecturers/professors on SD	Yes	Yes	Yes
Teaching across the natural sciences and social sciences faculties	No	Yes	No
Continuing education to the public on SD	Yes	Yes	Yes
<i>Research</i>			
SD institute research center	No	Yes	No
Funding for SD research	No	Yes	No
Promotion of holistic thinking in research	Yes	Yes	Yes
Use of research generated in SD teaching	Yes	Yes	Yes
Number of publications on SD	Yes	Yes	Yes
Patents in the field of SD	No	No	No
Creation of SD new knowledge and technologies	No	Yes	No
Research between the natural sciences and social sciences faculties	Yes	Yes	Yes
<i>Social and economic development</i>			
Exchange programs in SD	No	No	No
Joint SD degrees with other universities	No	No	No
Joint SD research with other universities	Yes	Yes	Yes
Collaboration in SD research projects with other universities	Yes	Yes	Yes
SD partnerships with other society stakeholders	No	Yes	No
Part of interdisciplinary SD expert networks	No	Yes	No
Academic staff involved in voluntary advisory activities in SD	Yes	Yes	Yes
SD events that are open to the public/community	Yes	Yes	Yes
Focus on social, environmental and economic sustainability	Yes	Yes	Yes
Employment opportunities	Yes	Yes	Yes

(continued)

Initiative for sustainable development	USTB	CUSIT	SUIT	Sustainability in higher education
<i>Awareness and volunteerism</i>				
SD working group with members from different departments	Yes	Yes	Yes	
Policies that promote SD for all students and staff	No	No	No	
Sustainable practices for students	Yes	Yes	Yes	
Student participation in SD activities	Yes	Yes	No	
Employees and other stakeholders act sustainably	Yes	Yes	Yes	
SD efforts are visible throughout the campus	Yes	Yes	Yes	
Staff and researchers are aware of SD	Yes	Yes	Yes	
Staff and researchers are engaged in SD research	Yes	Yes	Yes	
<i>Assessment and reporting</i>				
Assessment of SD issues	No	No	No	
Communication of SD activities	No	Yes	Yes	
Environmental reports	No	No	No	
Sustainability reports	No	No	No	
National environmental universities' rankings	No	No	No	
National SD universities rankings'	No	No	No	
International environmental universities' rankings	No	No	No	
International SD universities' rankings	No	No	No	
<i>Infrastructure</i>				
Campus operations	Yes	Yes	Yes	
Education	Yes	Yes	Yes	
Research	Yes	Yes	Yes	
Outreach	No	No	No	
On-campus experiences	Yes	Yes	Yes	
Assessment and reports	No	No	No	
Cultural development	Yes	Yes	Yes	

Notes: USTB = University of Science and Technology Bannu; CUSIT = City University of Science and Information Technology; and SUIT = Sarhad University of Science and Information Technology

Source: Authors compilation

Table 5.

operational documents, and departmental initiatives which could be directly linked to ESD. The data collection period started from January to September 2019.

Findings

Starting from the first section of [Table 5](#) these universities show a low level of implementation and reporting as mostly the top management has no written commitment on the integration of sustainability such as vision, mission, goals and objectives. Nevertheless, the top management of the sample universities expresses verbal commitments for sustainability and sustainable development. The findings are in line with the previous studies showing that integration of sustainable development programs is an issue of HEIs both in developed and developing countries ([Kapitulčinová et al., 2016](#); [Sassen and Azizi, 2017](#)).

On environmental sustainability, the sample universities are somehow committed to reducing the negative environmental footprints. Mostly, these universities are trying to reduce the use of energy. They have installed energy-efficient equipment such as energy savers, DC invertors for cooling, cooling plants and solar systems as a source of renewable energies. These universities are committed to the reduction of waste and disposals by

frequently sending notices, etc. to the stakeholders for waste reductions. They are also committed to using recycled materials such as the re-use of papers. Likewise, the CUSIT has announced paperless and reduce the use of paper policy. For that, they track the number of prints for individual users and seeks comments on the violation of these rules. The CUSIT has also implemented the City University Learning Management System for a fully paperless environment, which is used by the students, faculty and administrative staff for course delivery and other internal communications. USTB, as compared to the other two HEIs in the study, has shown good commitment toward the environmental aspect of sustainability. For instance, the university has recently mandated to bring or plant a tree or plant by each student who is graduated. Besides substantial landscaping, beautification and planting new lawns and gardens in the recent past, the university has also successfully shifted its admissions to the online system. About promoting a paperless environment, the university also provides many forms and other academic and research materials important for students on its website. The university is planning to purchase and implement a "Campus Management System" that could lead to a pure paperless organization by regulating its all activities like course registration, examinations and its related issues, result submission and announcement, file tracking, class attendance, students' record and payment for extra classes, medical and traveling and daily allowances, security and vehicle parking, among others. However, the university is yet to shift to solar energy as an alternative source of energy for light from the diesel-run generators in a remote area where the load shedding is severe and affects academic activities. Overall, environmental sustainability considers as one of the important dimensions of ESD (Ceulemans *et al.*, 2015), and these universities are keen to participate in it.

For the implementation of ESD in the HEIs, curricula are also considered one of the important aspects, among others (Alshuwaikhat *et al.*, 2016; Bothun, 2016). It has been observed that sample universities are relatively performing well in this area. Most of the sample universities are offering sustainability centric courses on under and postgrad levels. They are offering courses such as corporate sustainability, corporate social responsibility, business ethics and society, carbon emissions and management, ESD, environmental social and governance and some other related courses. Apart from that, these universities are guest lectures and international conferences on the topic of sustainable development and sustainability issues. Concerning curricula, the Institute of Management Sciences of the USTB has declared one credit hour CSR activities mandatory for all the students enrolled in Master of Science and Philosophy (MS/MPhil) of the institute.

Research in the area of ESD further enhances domains and paradigms of sustainability in the HEIs (Dlouhá and Pospíšilová, 2017). In research, these universities are committed to researching the subject areas. Among these universities, CUSIT has also established City University Center for Sustainability Studies (CUCSS), http://cusit.edu.pk/cusitnew/viewCentreofExcellence.php?coe_id=1 which is closely working on sustainability areas. The postgraduate students are assigned sustainability connected research projects. The student will get their degrees after completing these projects. Moreover, the concerned faculty members are also involved to publish relevant research in high index journals.

On the social and economic development front, the performance of universities under study is somewhat mild. In some aspects, they are performing well, whereas on the other fronts they lack proper link with other sister institutes under the umbrella of sustainable development. The universities under study have no such exchange programs and joint degrees with other institutes. While in the domains of research and collaboration, and academic level all three universities are involved with other sister institutions to promote sustainable development. In the case of social and economic sustainability, their

performance is worth noticing but on the environmental front again these universities are playing on a low sustainable level.

In the domain of awareness and volunteerism, all three universities are performing well. All the sample universities are involved in many activities to create and spread awareness regarding social, economic and, especially environmental sustainability by conducting seminars, social gatherings, and official meetings. These universities are somewhat involved in promoting sustainable policies and practices by involving students and staff to go green especially by plantations inside and outside the universities. Among these universities, SUIIT observes annual plantation days whereby different plants are distributed and planted in various parts of Peshawar city. These plants are donated by volunteers and faculty members of the university to promote a clean and green environment. Some faculty members in these universities are involved in research in the area of social, economic and environmental sustainability, whereas awareness agenda is promoted through newspaper articles and social media to attract volunteers. Regarding society, UTBSB arranges awareness seminars and workshops on AIDS and POLIO for the local community. However, these programs have no proper schedule and thus these should be streamlined. The university also has an active blood donor society that has a large number of members – teachers, staff and students. Overall, it could be concluded that the university is good at the on-campus operations, especially regarding the environmental aspect of sustainability but it is yet to mark any major success or impact on the societal and economic fronts of sustainability.

In the case of assessment, no real effort is apparent in all the three sister institutions. There is no proper assessment process and reporting of sustainable practices and development in these universities. There is a lack of official communication and reporting regarding environmental and sustainability practices inside and outside universities. Furthermore, there is no national and international environmental and sustainability-based ranking of these universities as there is no environment and SR.

The infrastructure of all these universities in terms of buildings, education and research activities is observed to be motivated toward sustainable practices and development. The overall official operations in these universities are somewhat inclined toward promoting sustainability. In terms of education, these universities have different courses related to economic, social and environmental sustainability at the postgraduate level. Research scholars and faculty members in Management, Social and Life sciences are involved in practical research on social, economic and environmental fronts of sustainability. Also, some sustainable practices are experienced inside the campus, but they are very low in number. The awareness efforts of these universities have resulted in the development of culture to practice sustainable activities at least inside universities. On the other hand, there is no reporting related to sustainability in these universities.

Conclusion and recommendations

This study aimed to conceptualize and develop a sustainability integration framework, based on prior relevant literature, for HEIs in a developing country like Pakistan. The study also intended to test the validity and effectiveness of the said framework in three different universities which included CUIT, SUIIT and USTB where the former two were private universities located in the provincial capital of KP – Peshawar, whereas the latter is a public sector university in a remote southern district of KP - Bannu. In short, this study developed and tested a new sustainability integration framework in a rarely focused area of research, i. e. HEIs and that too in the least developed area of KP, Pakistan where universities and other educational institutes, especially targeted by a decade long wave of terrorism and war against terror in the aftermath of 9/11. The population of the study is also important where it

is yet to fully understand or integrate and comply with sustainable business practices at the corporate level in developing countries like Pakistan. The collected data from the websites, internal publish reports and documents and interviews with the high-ups of these universities disclosed that all the three universities in the study, despite the differences in type or level of compliance, were somehow perusing the broader agenda of sustainable development. Furthermore, it was also noted that the concept of sustainable development is embedded at departmental levels, i.e. in the curriculum, teaching, research, seminars, workshops and conferences which might be the outcome of the presence of some faculty members in these universities who earned their PhD degrees in the subject from abroad. However, the findings also provide evidence in that the integration, implementation, assessment and reporting aspects of the sustainability were very low and weak in all the sample universities of the study. This might be due to lack of awareness, assessment and reporting mechanism, budget, training, staff and will or focus at the top. The cross-comparison of the sample universities shows that private universities, as compared to the public, are relatively more inclined towards sustainable development. This may be due to the status and structure of the private universities or the sense of ownership which facilitates these universities in quick decision-making along with earlier and good compliance to new trends, reforms and regulations. A possible meaning to the findings could also be that universities in main or big cities are more inclined toward sustainable development than those located in remote areas. Furthermore, sustainable development in HEIs requires the allocation of ample resources which is comparatively easy in private universities than the public universities where a long and cumbersome process is involved for getting approvals and resources. Together, all these findings of the study imply that the weakness or variation in the level of compliance among universities may be the output of lack of awareness or policy and pressure from the top, among others. Hence, the regulators, HEC, and all other key stakeholders should set together to formulate and implement an effective strategy that could ensure the integration, implementation and reporting of a uniform index for sustainable development in all HEIs, whether private or public and located either in big cities or remote areas.

Overall, this study has both practical and theoretical contributions. The study conceptualized and proposed a unique sustainability integration framework for the HEIs of a developing country like Pakistan. Second, the study tested the framework in two private universities of a relatively developed city of Peshawar (in the context of KP province) and one public university of a remote area of KP. Besides, the study also provides important insights and practical implications for academia, regulatory bodies and universities to integrate, implement and report sustainability in HEIs. The findings show that somehow compliance exists at a low level or on the ground and from the bottom instead of from the top; hence, these efforts not only lack uniformity but also exhibit weakness in implementation with true spirit. Therefore, the interest and focus along with the proper allocation of resources from the top could increase the level of compliance and uniformity of these practices, among others. The study also offers theoretical contributions as the findings of the study add new insights to enrich the extant prior literature of sustainability in the HEIs particularly in the context of developing countries. Likewise, the study proves the applicability of stakeholders' theory in the HEIs to serve or satisfy the demands of different stakeholders and get the license to operate in society.

Limitations of the study

The study is not free from limitations. The results of the study are mainly based on the descriptive findings, hence, studies in future studies may focus on the quantitative aspects based on the newly developed framework. Likewise, this is the first-ever test of the newly

developed framework on only three universities in KP. Therefore, more and more studies may be carried out in the future to check the validity of the framework on a substantial number of both the public and private universities from the rest of the three provinces, i.e. Punjab, Sindh and Baluchistan as well. Future studies might also consider the qualitative aspect of the subject for a richer and clearer picture. Finally, future studies may also consider the scenario of developing and developed countries for the ESD in the HEIs.

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